

Learning path: from CS to Kubernetes

Broad map, not a syllabus. Order follows the instructor: CS, network, cybersecurity, system administration, cloud computing, DevOps, then Kubernetes. One heading at a time.

How we work

You ask "what do I learn now?". We open a single heading and apply it (internship style). No lecture dump until that heading is open. The Library app (API + UI on Minikube) is the spine from DevOps/Kubernetes onward. Earlier phases use small labs.

Phases (instructor order)

0 - Computer science (foundations)

OS ideas: process, memory, files, CPU vs I/O. Enough programming comfort to read scripts and APIs.

Done when: You can explain what a process, a file, and a port are without mixing them up.

1 - Network

TCP/IP, DNS, HTTP/HTTPS, ports, routing vs switching at a practical level, firewalls.

Done when: You can trace a request from browser to a server (name, IP, port, TLS).

2 - Cybersecurity

CIA triad, identity and access, least privilege, TLS as a concept, common failure modes (weak secrets, open ports, over-permission).

Done when: You default to who can do what and what is secret before you deploy anything.

3 - System administration

Linux daily: users, permissions, systemd/services, packages, logs, SSH, disk, networking on the box.

Done when: You can run and debug a service on a Linux machine without a GUI.

4 - Cloud computing

IaaS vs PaaS vs SaaS, regions, VMs, IAM, object storage, one provider at a glance (AWS or Azure or GCP).

Done when: You can place a VM and a network in a cloud and say what you are paying for.

5 - DevOps

Git, CI/CD idea, containers (Docker: images, Compose, ship), infra as code as a concept.

Done when: Library (or a small app) builds as an image and runs with Compose; a pipeline idea is clear.

6 - Kubernetes

Apps on a cluster, then storage, Helm, operate, GitOps, cluster security, cluster admin, one managed cloud.

Done when: You treat the cluster as how the app runs in production, not as a YAML museum.

Kubernetes (phase 6) headings only

6A Kubernetes apps. 6B Workloads and storage. 6C Helm. 6D Operate. 6E Delivery (CI + GitOps). 6F Security on the cluster. 6G Clusters (kind/kubeadm). 6H Managed Kubernetes (AKS or EKS or GKE). Later: service mesh, operators, Terraform-everything, CKS cram.

Current heading

Phase 0 - Computer science foundations. Closed until you say we start.